

GOVERNMENT OF INDIA — MARITIME ENVIRONMENTAL PROTECTION
DIVISION

HEADQUARTERS INDIAN COAST GUARD • NATIONAL TECHNICAL
RESEARCH ORGANISATION (NTRO)

TECHNICAL EVIDENCE DOSSIER & STATUTORY NOTICE OF VIOLATION
ISSUED UNDER SECTION 356E, MERCHANT SHIPPING ACT 1958 & MARPOL 73/78 ANNEX I

Dossier Ref:	ICG-MEP-DOSSIER-20260905-MUMB	Date of Issue:	05 Sep 2026, 16:04 UTC
Maritime Sector:	Offshore Maharashtra / Western EEZ	Legal Classification:	ILLEGAL DISCHARGE (CRIMINAL)
Target Vessel:	SAMUDRA SEVAK	IMO / MMSI:	9187766 / 419088112
Flag State:	India	Attribution Certainty:	64.6% (CONFIRMED)

1. SATELLITE RADAR REMOTE SENSING (SAR) OBSERVATION

Sensor Platform:	Sentinel-1B C-Band SAR	Acquisition Timestamp:	2026-09-04 01:25:00 UTC
Swath Mode & Pol:	Interferometric Wide (IW) Swath / VV	Ground Resolution:	10.0 meters/pixel
Observed Slick Centroid:	19.3588° N, 71.2811° E	Total Spill Surface Area:	24.46 km²
Estimated Discharge Mass:	538.0 Metric Tonnes	Lookalike Rejection Score:	80.0% Mineral Crude Confidence
ADIOS Weathering State:	Heavy Chocolate Mousse (Highly Viscous Emulsion)	Evaporative Loss / Mousse:	22.3% Evaporated / 75.0% Water Uptake

2. REVERSE LAGRANGIAN HYDRODYNAMIC DRIFT ANALYSIS (ORIGIN ATTRIBUTION)

Using an advection-diffusion Lagrangian numerical formulation integrated with real-time oceanographic current vector fields ($\Delta u=0.22$ m/s, $\Delta v=-0.12$ m/s) and Ekman surface windage (3.2% Stokes drift factor with 15° Coriolis deflection), the slick plume trajectory was hindcasted backwards in time. The particle convergence analysis resolved the **exact discharge origin point**:

Spill Origin Point (x_0, y_0):	19.46488° N, 71.24426° E	Calculated Slick Age:	8.5 Hours prior to SAR pass
Estimated Discharge Time:	T – 8.5h (05 Sep 2026 approx 02:40 IST)	Total Hydrodynamic Drift:	10.32 km displacement

3. AIS CORRELATION & KINEMATIC BEHAVIORAL ANOMALIES

Analysis of **3 vessels** operating in the sector filtered down to **3 corridor candidates**. Vessel **SAMUDRA SEVAK** exhibited definitive spatio-temporal intersection with the discharge epicenter:

Evaluation Metric	Observed Telemetry Value	Anomaly Assessment	Score Weight
Closest Point of Approach (CPA)	1.12 NM (2.07 km)	Exact spatial co-location with slick root	82.4/100
Temporal Coincidence	$\Delta t = 0.00$ hours from origin	Precise temporal synchronization at t_0	100.0/100
Speed Over Ground (SOG)	Drop: 7.5 → 7.5 kts	Steady cruising speed maintained	30.0/100
Vessel Risk Category	Offshore Supply Vessel (DWT: 4,100 T)	High capacity oily water / slop separator	30.0/100

4. NON-COOPERATIVE / DARK VESSEL RADAR SURVEILLANCE AUDIT

TACTICAL RADAR TARGET INTERCEPTION: CFAR point-target detection extracted **6 radar contacts** with metallic corner-reflector signatures ($>+5$ dB σ^0) that broadcast **ZERO AIS transponder telemetry**.

Radar Target ID	Position (Lat, Lon)	RCS (σ^0 dB) / Length	Threat Classification
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RADAR-TGT-859	19.3141° N, 71.2707° E	27.2 dB / ~200.0m	HIGH PRIORITY SUSPECT (COVERT DISCHARGE)
RADAR-TGT-1491	19.3271° N, 71.4407° E	28.3 dB / ~150.0m	UNREGISTERED MARITIME CONTACT
RADAR-TGT-3149	19.3594° N, 71.2250° E	27.2 dB / ~250.0m	HIGH PRIORITY SUSPECT (COVERT DISCHARGE)

5. STATUTORY DIRECTIVES & DETENTION ORDER

NOTICE TO MASTER, OWNER, AND FLAG ADMINISTRATION:

Pursuant to powers vested under Section 356J and Section 356E of the Merchant Shipping Act, 1958:

- 1. **PORT STATE CONTROL INTERCEPTION:** The Principal Officer, Mercantile Marine Department (MMD), is directed to **DETAIN** vessel **SAMUDRA SEVAK (IMO: 9187766)** upon entry into any Indian port or anchorage until mandatory oil record book (ORB Part I & II) inspection and bilge manifold swab sampling are concluded.
- 2. **FINANCIAL SECURITY:** A maritime environmental lien of **INR 25,00,00,000 (Twenty-Five Crore Rupees)** is hereby levied to cover offshore containment, dispersant spraying, and coastal remediation costs.
- 3. **FLAG STATE NOTIFICATION:** A copy of this satellite radar and AIS kinematic forensic package is dispatched to the International Maritime Organization (IMO) and the Maritime Administration of India.

Forensic Chain-of-Custody SHA-256 Hash:
3062dbfecb37d1628eaf48f1e55dc4226e1b5a3257939095ca09624580340584

Authorized Signatory:
Inspector General (Operations & MEP)
Indian Coast Guard Headquarters, New Delhi